FACULTY OF ENGINEERING AND TECHNOLOGY

Electrical and Computer Engineering Department

Artificial Intelligent

ENCS3340

Project #1 Report

Optimization Strategies for Local Package Delivery Operations

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Section: 1

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Part A: Simulated annealing

- **Problem formulation:**

- **Initial state:**
  Randomly distribute packages into vehicles under conditions
  1- not exceed the vehicle capacity
  2- drop the lowest priority package in Case - Overcapacity Package
     this done in generate_initial_state() function
- **Next state:**
  Two cases
  1- only one vehicle: just swap two packages in its route
  2- more than one vehicle: swap one package from one vehicle to another under the constrains of the capacity
     this done in generate_next(state) function
- **Objective function:**
  Based on:
  1- Total distance: calculate the total traveled distance by all the vehicles in the shop using Euclidean distance formula.
     This done in total_distance(state) & euclidean_distance(point1, point2) functions
  2- priority delivering: give a reward score value as big as the height's priority pkgs delivered first in state routes. This don in priority_score(state)
     The objective_function(state) function returns total distance-score, which we try to minimize to reach the best solution.
- **Solution:**
  The solution is a state of vehicles and the pkgs order inside each.

- **Algorithm:**
  It explores neighbors(next)of current state and sometimes accepting worse solutions based on prob to avoid getting stuck in local minima.
  If new solution is **better** → always accept it.
  If it's **worse** → accept it with probability exp (-delta / T):
  High T → more likely to accept worse solutions.
  Low T → behaves more like greedy search.
  As you see the effect parameter as **T** "Initial Temperature"
  Cooling Rate (cooling_rate): Controls how fast the temperature drops
  Stopping Temperature (stopping_temp): The algorithm stops when T drops below this value.
  This means the lower value = the longer search and more possibly get better solution.
  Iterations per Temperature (iterations_per_temp): Number of neighbor (next)attempts at each temperature level. This means the more iterations →the better exploration but in contrast slower runtime.

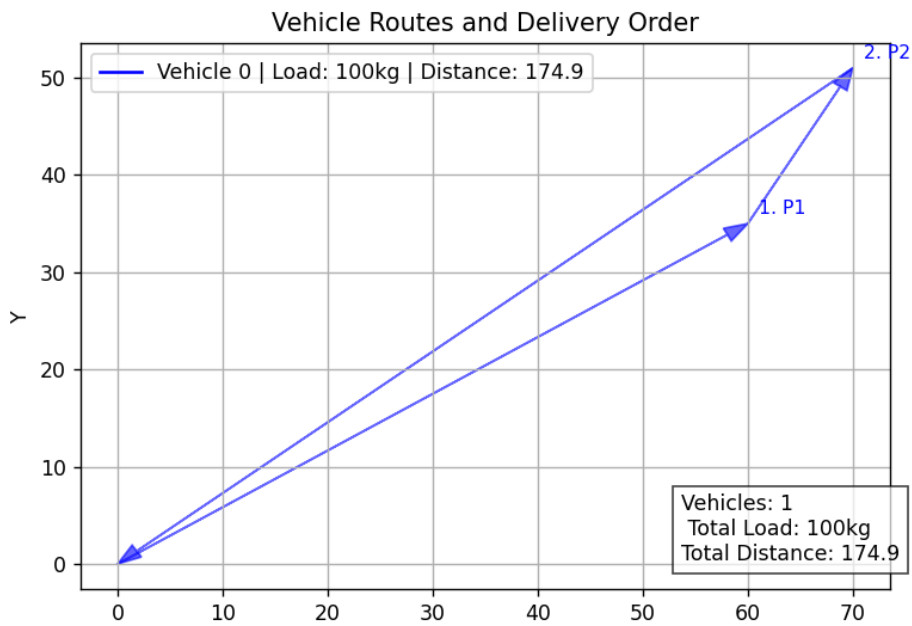
➢ Test cases:

1- Priority Handling Test:

- **Input**: 1 vehicle with a capacity of 100 kg, and 3 pkgs each 50kg with 3,1,2 priorities and different locations.

```
v0 100
#package x y waight priority
p0 10 10 50 3
p1 60 35 50 1
p2 70 51 50 2
```

- **Output:**



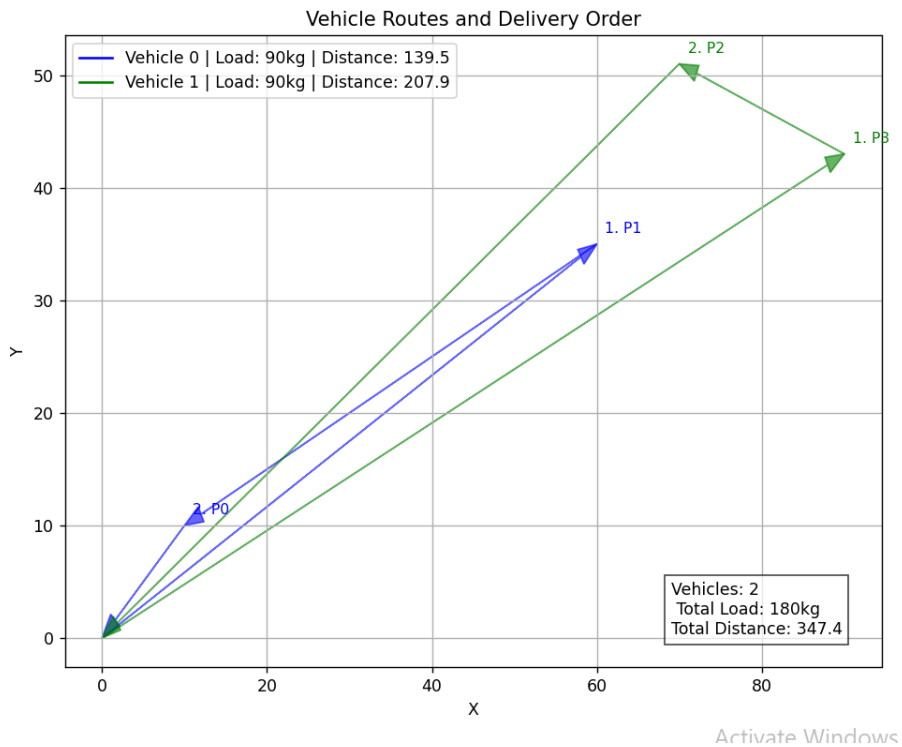

```
packeges {0: [10, 10, 50, 3], 1: [60, 35, 50, 1], 2: [70, 51, 50, 2]}
number of packeges is : 3
vehicles:  [100]
number of vehicles is : 1
 Could not place package 1 due to weight constraints.Dropping package with lowest priority.
Swapped out package 0 (priority 3) to place 1 (priority 1) in vehicle 0
The initial state is  {0: [2, 1]}
Total Distance for initial state: 174.94

Best vehicle assignments:
Vehicle 0 (Capacity: 100, Load: 100): Packages [1, 2]

Total Distance: 174.94
```

2-

```
v0 100
v2 100
#package x y waight priority
p0 10 10 30 3
p1 60 35 60 1
p2 70 51 50 2
p3 90 43 40 2
```




```
packeges {0: [10, 10, 30, 3], 1: [60, 35, 60, 1], 2: [70, 51, 50, 2], 3: [90, 43, 40, 2]}
number of packeges is : 4
vehicles:  [100, 100]
number of vehicles is : 2
The initial state is  {0: [1, 0], 1: [3, 2]}
Total Distance for initial state: 347.40

Best vehicle assignments:
Vehicle 0 (Capacity: 100, Load: 90): Packages [1, 0]
Vehicle 1 (Capacity: 100, Load: 90): Packages [3, 2]

Total Distance: 347.40
```

3- Overcapacity Package:

```
v0 100
v2 100
#package x y waight priority
p0 10 10 150 1
```

last

```
C:\Users\User\PycharmProjects\osproj\.venv\Scripts\python.exe C:\Users\User\PycharmProjects\osproj\last.py
packeges {0: [10, 10, 150, 1]}
number of packeges is : 1
vehicles:  [100, 100]
number of vehicles is : 2
 Could not place package 0 due to weight constraints.Dropping package with lowest priority.
Could not place package 0 (priority 1) even after attempting swaps. Dropping it.
no more package to deleveir! exit

Process finished with exit code 0
```

Part B: Genetic algorithm

- **Problem formulation:**

- **Goal:**
  Assign packages to vehicles and determine the sequence of deliveries such that:
  1. The total distance traveled by all vehicles is minimized.
  2. High-priority packages are delivered earlier in the route, earning a priority reward

- **Objective Function:**

  The algorithm minimizes a cost function defined as:

$$\text{Objective Value} = \text{Total Distance} - \text{Priority Reward}$$

  Total Distance: The round-trip distance each vehicle travels, computed using the Euclidean distance between package locations and the shop.

  Priority Reward: Packages delivered earlier in a vehicle's route contribute more to the reward. Higher priority packages (lower priority number) give more reward.

- **Genetic Algorithm:**

1. Initialization:

- Generate random valid assignments of packages to vehicles.
- Ensure no vehicle exceeds its weight capacity.
- Skip unassignable packages with appropriate warnings.

2. Selection:

- Select parent solutions using Fitness proportionate selection, where fitness is inversely proportional to the objective value.

3. Crossover:

- Swap the assignments of common packages between two parents.
- Only accept children that maintain capacity constraints.

4. Mutation:

- With a small probability, randomly reassign packages between vehicles while ensuring capacities are not violated.

5. Fitness Evaluation:

- Evaluate each solution using the objective function.
- Track and update the best solution found across generations.

- **Termination Criteria:**

  The algorithm runs for a fixed number of generations (default: 500). The best solution and the generation it was found in are returned.

➢ Test cases:

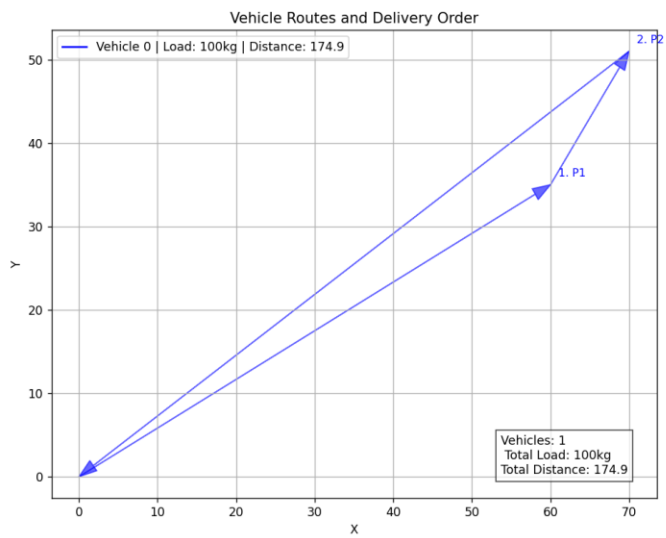
1- Priority Handling Test:

- **<u>Input</u>**: 1 vehicle with a capacity of 100 kg, and 3 pkgs each 50kg with 3,1,2 priorities and different locations.

```
v0 100

p0 10 10 50 3
p1 60 35 50 1
p2 70 51 50 2
```

- **<u>Output</u>:**



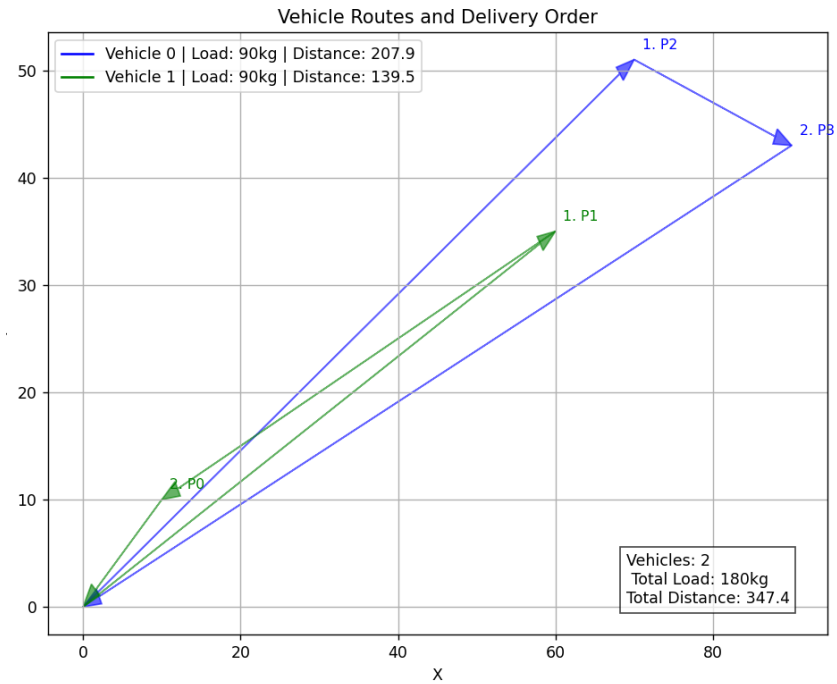

```
Vehicles: [100]
Number of vehicles is: 1
Packages: {0: [10, 10, 50, 3], 1: [60, 35, 50, 1], 2: [70, 51, 50, 2]}
Number of packages is: 3
Which algorithm to use to generate the solution:
1- Simulated annealing algorithm.
2- Genetic algorithm.
2
Warning: Some packages were dropped due to capacity limits.
 - Dropped package 0 (priority 3, weight 50)

Best vehicle-package assignment:
Vehicle 0 (Capacity: 100, Load: 100): Packages [1, 2]
Total Distance: 174.94
Best solution found in Generation 1
```

2-

```
v0 100
v1 100

p0 10 10 30 3
p1 60 35 60 1
p2 70 51 50 2
p3 90 43 40 2
```




```
Vehicles: [100, 100]
Number of vehicles is: 2
Packages: {0: [10, 10, 30, 3], 1: [60, 35, 60, 1], 2: [70, 51, 50, 2], 3: [90, 43, 40, 2]}
Number of packages is: 4
Which algorithm to use to generate the solution:
1- Simulated annealing algorithm.
2- Genetic algorithm.
2

Best vehicle assignments:
Vehicle 0 (Capacity: 100, Load: 90): Packages [2, 3]
Vehicle 1 (Capacity: 100, Load: 90): Packages [1, 0]
Total Distance: 347.40
Best solution found in Generation 1
```

3- Overcapacity Package:

```
v0 100
v1 100

p0 10 10 150 1
```

```
main

C:\Users\HP\PycharmProjects\Ai_Project1\.venv1\Scripts\python.
Vehicles: [100, 100]
Number of vehicles is: 2
Packages: {0: [10, 10, 150, 1]}
Number of packages is: 1
Which algorithm to use to generate the solution:
1- Simulated annealing algorithm.
2- Genetic algorithm.
2
Warning: Some packages were dropped due to capacity limits.
 - Dropped package 0 (priority 1, weight 150)
No valid initial population generated. Skipping evolution.
No valid solution found.
```